# Tumor Hypoxia and Lung Cancer Survival

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### Research question

Are gene-expression-derived hypoxia scores associated with overall survival in lung adenocarcinoma after adjustment for clinical characteristics?

### Data and methods

TCGA-LUAD PanCancer Atlas clinical records and Buffa, Ragnum and Winter hypoxia scores. Cohort flow: 566 initial patients, 485 complete analysis cases, including 177 deaths.

Kaplan–Meier estimation, log-rank comparisons, multivariable Cox models, proportional-hazards diagnostics and sex-by-stage stratification.

### Selected results

- Overall survival estimates at 12, 36 and 60 months: 0.875, 0.615 and 0.392.
- In the reported Cox model, a 10-unit Buffa-score increase has HR 1.117 (95% CI 1.036–1.206).
- Historical global Schoenfeld tests give $p = 0.016$ before and $p = 0.461$ after stratification.

### Interpretation and limitations

These are observational associations, not clinical decision rules. Hypoxia signatures are indirect measures, subgroup sizes are limited, and some detailed diagnostic plots still need historical cross-checks.

### Reproduce

```
Rscript scripts/run_all.R
```

Repository and full documentation